

Numerical Experiments for Lévy Score-Corrected Compound-Poisson Equilibrium Sampling

Abstract

We evaluate Lévy score-corrected compound-Poisson (LSC-CP) sampling on four multimodal Boltzmann model systems: a one-dimensional double well, a planar forty-mode Gaussian mixture, a depth-balanced three-well Müller–Brown landscape embedded in ten dimensions, and a coupled ϕ^4 chain in twenty-four dimensions. Across the reported horizons, LSC-CP markedly reduces terminal basin-population error relative to local samplers and, in E2–E4, relative to raw compound-Poisson dynamics; it also yields nonzero post-settling effective sample sizes for every basin indicator. E1 illustrates why basin occupancy alone is insufficient: raw CP happens to recover the two basin masses while still distorting tail and energy statistics. With the current MoG40 jump interval [10, 14], its sliced W_2 lies within the finite-sample reference band, whereas its MMD remains marginally above that band. Thus, agreement of basin populations or one projection-based metric does not imply agreement of the complete distribution. The implementation uses finite chord quadrature, tamed Euler updates, an exponent cap, and projection onto a finite computational box. These practical stabilizations control overflow and large updates and improve numerical robustness in the stiff score regime, but they define an approximate finite-step kernel. Potential-energy discrepancies that are large relative to the chain-mean MCSE, conditional on the frozen reference estimate, are therefore reported explicitly. The evidence supports accelerated interbasin exploration and accurate population recovery with observable-dependent numerical bias, rather than exact finite-step Boltzmann sampling. No physical kinetics is inferred.

1 Overview and compared methods

All experiments target a Boltzmann law

$$\pi_\beta(x) = Z_\beta^{-1} e^{-\beta V(x)}, \quad x \in \mathbb{R}^d. \quad (1)$$

The sampler of interest augments overdamped Langevin dynamics with a finite compound-Poisson jump process of measure $\nu(dr) = \lambda\rho(dr)$ and adds the stationary Lévy-score drift $S_{\nu,\beta}$ so that the combined generator

$$A_{\nu,\beta}f = [-\nabla V + S_{\nu,\beta}] \cdot \nabla f + \beta^{-1} \Delta f + J_\nu f \quad (2)$$

is weakly stationary for π_β (Appendix A). The practical estimators evaluate the score integral by deterministic chord quadrature where affordable and by *paired* random-bank estimators in higher dimension: the realized jump displacements and the Lévy-score drift of one step are generated from the same random finite measure. This pairing aligns the drift and jump measures before the remaining finite chord quadrature is applied (Appendix A.1). The reported variants are: deterministic shell quadrature (double well), full numerical chord–radius–angle quadrature for the annular MoG40 law, the paired single-atom estimator LSC-CP-RA (double well and MoG40, as an estimator-agreement check), deterministic shell quadrature in the two higher-dimensional systems, and the

paired multi-atom estimator LSC-CP-MA (Müller–Brown and ϕ^4). The numerical artifacts store the latter under the key LSC-CP-MA; legacy plotting aliases call it LSC-CP-RA (4) or LSC-CP-RA (8). We use MA throughout this note because the implementation draws one displacement from every atom family, distinguishing it from the single-displacement RA estimator in E1–E2.

Baselines are the unadjusted Langevin algorithm (ULA), Metropolis adjusted Langevin (MALA), the BAOAB underdamped integrator (BAOAB), an α -stable Lévy-flight analogue (FLA), and parallel tempering (PT) with an automatically tuned ladder and fully counted replica cost. Raw compound Poisson (CP), i.e. the same jump bank *without* the score correction, is included as a stationarity-defect ablation: its invariant law is not π_β , so it is a control for the value of the correction, not a fair efficiency baseline. All methods share per-seed initial ensembles. In the double-well and MoG40 full-law experiments, CP and deterministic/numerical-quadrature LSC-CP additionally share the jump random stream pathwise. In the Müller–Brown and ϕ^4 experiments, raw full-law CP and LSC-CP-MA match the marginal jump geometry, weights, and intensity, but do not use the same realized random banks and hence are not pathwise paired.

The implemented scheme uses finite chord quadrature, log-domain accumulation with an exponent cap, tamed Euler drift updates, and projection onto a finite computational domain to stabilize evaluation in the stiff-score regime. These operations define the finite-step algorithm evaluated below. The stationarity identity applies to the ideal continuous-time generator with the exact, uncapped Lévy-score field on an unbounded state space. We assess the equilibrium fidelity of the discretization through timestep and quadrature sensitivity checks, observable biases, and cap/projection activation rates. Passage statistics are interpreted solely as algorithmic mixing diagnostics; no physical reaction kinetics is inferred.

2 Benchmark systems

	Double well	MoG40	Depth-balanced MB	Coupled ϕ^4
dimension d	1	2	10	24
β	8	8	24	8
particles N	16,000	2,500	2,000	1,000
horizon T	100	100	200	100
Δt (used)	0.005	0.010	0.005	0.002
steps	20,000	10,000	40,000	50,000
seeds	24	24	16	16
jump rate λ	1	1	1	1
score quadrature	$q_\theta=16, q_\rho=4$	$q_\theta=16, q_\rho=4, m_\phi=16$	$q_\theta=8, q_\rho=4$	$q_\theta=32, q_\rho=4$
score estimator	deterministic + RA	numerical annulus + RA	deterministic + MA	deterministic + MA
weak-identity residual	1.5×10^{-7}	2.1×10^{-8}	5.0×10^{-7}	2.0×10^{-7}

Table 1: Reported numerical configurations and uncapped weak drift–jump identity residuals (diagnostic tolerance 10^{-6} ; Appendix E). The residual measures generator-level drift–jump cancellation for a finite family of test functions. Here N is the number of particles per seed block; a batched run with S seeds evolves SN particles and computes metrics separately on each N -particle block. Full model, bank, and ladder specifications are in Appendix B.

Artifact–source distinction for MoG40. Table 1 records the configuration that generated the saved results: $[10, 14]$ and $m_\phi = 16$. The current experiment builder retains $[10, 14]$ but raises the default angular order to $m_\phi = 64$ for a future rerun. Because no sampler was rerun for this revision,

all numerical tables remain aligned with the saved $m_\phi = 16$ artifacts and must not be attributed to the newer default. The saved E2 manifest does not serialize the annulus endpoints or resolved estimator. Its recorded repository SHA resolves to the older [4, 15] law with analytic **MoG40Score**, whereas the jump-term certificate ratios and evaluation ledger identify the executed configuration as [10, 14] with numerical **ShellScore** at $m_\phi = 16$. Thus the executed law and estimator are identifiable from numerical fingerprints, but the exact executed source-tree provenance is not recoverable from the archived metadata. A future production run should serialize both the requested and resolved configurations.

Double well ($d = 1$). $V(x) = (x^2 - 1)^2$ at $\beta = 8$: two equal wells at ± 1 separated by a dimensionless barrier $\beta\Delta V = 8$. The jump bank is a symmetric shell at ± 2 with radial half-width 0.2. This is the transparent two-state test of target recovery and interwell communication. Every main relaxation run starts from

$$X_0 = -1 + 0.05\xi, \quad \xi \sim N(0, 1). \quad (3)$$

MoG40 ($d = 2$). Using NumPy’s PCG64 generator with seed zero, the centers are drawn once as $\mu_k \sim \text{Unif}([-40, 40]^2)$ and then frozen (the code call is `default_rng(0)`). The target and potential are

$$\pi_\beta = \frac{1}{40} \sum_{k=0}^{39} N(\mu_k, I_2), \quad V(x) = -\beta^{-1} \log \sum_{k=0}^{39} e^{-\|x - \mu_k\|^2/2}. \quad (4)$$

The main relaxation ensemble is $X_0 = \mu_0 + 0.5\xi$, $\xi \sim N(0, I_2)$, and basin labels are assigned by the nearest center. The jump law is deliberately generic: radius uniform on [10, 14] and direction uniform on the circle. The interval is narrower than the former [4, 15] rule, while its upper radius retains a connected mode graph under both recorded mode-hit thresholds. Neither the annulus law nor **ShellScore** receives the mode list as a separate geometry-design input; the numerical score accesses the centers only through black-box evaluations of V . Exact-reference sampling, nearest-center labels, and the connectivity check use the centers directly. This is the many-basin stress test.

Depth-balanced Müller–Brown ($d = 10$). A Müller–Brown-type three-well landscape whose well depths are retuned to be equal, evaluated in a two-dimensional latent plane and embedded in ten dimensions by a fixed linear map with Gaussian auxiliary coordinates ($\sigma_{\text{aux}} = 0.4$), at $\beta = 24$. The equal depths separate multimodality from metastability: the channels retain dimensionless barriers $\beta b_{AB} \simeq 11.1$ and $\beta b_{BC} \simeq 4.0$ while the equilibrium masses stay comparable. The bank has four directed relay atoms $A \leftrightarrow B \leftrightarrow C$ (no direct A – C atom), so transport must use the relay through B . This tests curved, nonuniform basins and a designed jump graph. If z_C is the refined latent minimum of well C , the main initial law is

$$z_0 = (z_C, 0_8) + 0.05\xi, \quad X_0 = z_0 B^\top, \quad \xi \sim N(0, I_{10}). \quad (5)$$

Coupled ϕ^4 chain ($d = 24$). Twelve periodic two-component sites, $V(q) = \frac{\kappa}{2\delta} \sum_i \|q_{i+1} - q_i\|^2 + \delta \sum_i W(q_i)$ with $\kappa = 2.5$, $\delta = 1/12$, $\beta = 8$, and a tilted double-double-well site potential W (Appendix B). The target has four coherent phases; the bank has eight directed edge atoms along the phase square (diagonals excluded because their coherent chords cross the field-zero hilltop). The primary collective variable is the spatial mean $\bar{q} = \frac{1}{12} \sum_i q_i \in \mathbb{R}^2$. This is the many-body

example: a collective transition in a coupled field. The main relaxation runs start from the coherent minus-minus phase,

$$Q_0 = m_{--} + 0.05 \xi, \quad \xi \sim N(0, I_{24}). \quad (6)$$

The four systems form a hierarchy of equilibrium questions—two-state balance, many-mode occupancy, curved-basin relay transport, and a collective field transition—rather than four independent leaderboard tasks.

3 Protocol, references, and numerical parameter selection

Run protocol. For each system all seeds are batched into one ensemble per method with per-seed initial blocks; every method starts from the identical localized per-seed ensemble. These labels denote blocks within one batched method run, not independently launched jobs; the dynamics use one method-level random generator, and batch wall time and NFE are repeated in the seed rows. Consequently, the reported seed standard deviation describes block-to-block variation within that batch rather than run-to-run timing uncertainty. We used dyadic timestep sensitivity checks based on terminal W_2 , basin TV, MMD, and explored-mode-coverage diagnostics for the π_β -targeting methods. The non-targeting FLA and raw CP chains were recorded but did not determine the timestep. Score quadrature was selected analogously, with the additional uncapped weak-identity residual. These checks guided numerical parameter selection, but they did not include post-settling mean energy or collective-variable bias and therefore do not establish convergence of the invariant measure of the stabilized chain. The saved refinement helper selects the first method whose key begins with LSC-CP; in E3 and E4 this is the deterministic-quadrature arm. Accordingly, the quadrature-refinement and π -start hold gates do not separately validate the paired MA arm, which is assessed only through its terminal and post-settling diagnostics.

The explored-mode-coverage (EMC) diagnostic is higher-is-better, whereas the current automated agreement helper is written for lower-is-better errors. Consequently, EMC is recorded in the refinement tables but does not provide a valid acceptance gate in the saved runs. The refinement evidence should therefore be read as coming from W_2 , basin TV, MMD, and the weak-identity residual. A corrected implementation should gate on $1 - \text{EMC}$.

References and finite-sample levels. The double well uses a dense-grid inverse-CDF reference; MoG40 an exact mixture draw; Müller-Brown a fine two-dimensional latent grid combined with exact Gaussian auxiliaries; ϕ^4 a Laplace-mixture proposal corrected by self-normalized importance weights (direct SNIS for basin masses, energy moments and the energy-overlap histogram, CV means, and the free-energy surface; finite sampling-importance-resampling only where unweighted clouds are required). The ϕ^4 weight diagnostics for the reference are: 2×10^5 proposals, weight ESS 41,182 (fraction 0.206), maximum normalized weight 3.6×10^{-3} (Appendix D). A separate $t = 300$ parallel-tempering check gave phase masses (0.365, 0.204, 0.234, 0.197), whose TV distance from the SNIS masses is 0.042. Consequently, differences below this scale in the ϕ^4 basin metrics are reference sensitive. For W_2 , MMD, basin TV, EMC, E1 density TV, and E3 ten-dimensional sliced W_2 , a finite-sample reference level is estimated from 20 independent reference replicates at matched sample size. For lower-is-better errors, “within the reference band” means below mean + 3 sd. No such band is currently computed for FES RMSE, basin KL, energy-moment errors, energy-overlap deficit, KSD, or the archived first-CV CDF/KDE discrepancies; those quantities must not be described as being “at the floor.”

Terminal-sample and reference-density figures. The method-sample panels were reconstructed from the saved `positions.csv` files; the reference-density panels were evaluated separately from the target density or reference integral described below. No sampler trajectory was rerun. Each double-well method is shown with the normalized target density $p(x) \propto \exp[-8(x^2 - 1)^2]$ and a deterministic display subset of its saved samples; the panel text contains neither density qualifiers nor point counts. For E2–E4, each method has one terminal scatter plot and each experiment has one separate reference-density plot. E2 evaluates the exact forty-component Gaussian-mixture density, and E3 evaluates $p(z) \propto \exp[-24V_{\text{MB3}}(z)]$ in the active plane. The E4 visualization uses an independent, fixed-seed CPU direct-SNIS construction followed by a phase-conditioned weighted KDE of $\bar{q}$; it is not an analytic marginal and is distinct from the archived GPU SNIS cloud used for numerical metrics (Appendix D). Plot titles contain only the method name or “Reference.”

Stability diagnostics. We monitored nonfinite states, score-cap and state-box projection frequencies, jump-boundary contact per applied jump, and basin-map-domain escape. The uncapped weak-identity residual was evaluated on a domain extending at least one maximum jump beyond the high-probability region. The measured values and domain-coverage rationale appear in Appendix E. These quantities diagnose numerical robustness and exposure to the stabilizations; small activation frequencies do not prove that the finite-step kernel has invariant law π_β .

4 Equilibrium recovery

Four observations organize Table 2 and Figure 1.

(i) *Local dynamics remain strongly metastable.* On every system, the terminal ULA, MALA, and BAOAB ensembles remain strongly localized relative to the target (basin TV 0.31–0.93, basin KL 1.1–4.2). Occasional label changes occur in some of the post-settling MoG40 and Müller–Brown traces, but they are too sparse to recover all basin populations. This quantifies the metastability that the nonlocal methods are intended to address.

(ii) *LSC-CP improves basin recovery, with metric-dependent residual error.* For the reported LSC estimator in each system, both the main W_2 and basin TV lie within their finite-sample reference bands. In MoG40, the numerical values are $W_2 = 2.05$ versus an upper reference level 2.142, and basin TV 0.053 versus 0.069. Its MMD, 0.03199, is slightly above the corresponding upper level 0.03171, so the full two-dimensional discrepancy has not been shown to be indistinguishable from reference sampling noise. Basin KL is small for the LSC estimator in every system, but no KL reference band was computed. In ϕ^4 , differences in basin metrics below 0.042 remain sensitive to the independent-reference discrepancy described in Section 3.

The updated, uniformly weighted augmented-grid FES metric changes the earlier ordering. Parallel tempering has lower FES RMSE than LSC-CP on the double well (0.393 versus 1.210), MB (0.616 versus 1.053 for the multi-atom estimator), and ϕ^4 (1.063 versus 1.294). On MoG40 the corresponding values are 0.574 and 0.695. Thus LSC-CP improves the low-dimensional FES relative to local and raw-jump dynamics, but it is not the most accurate method under this FES definition. The α -stable flight can have a small value for an individual terminal metric, but its invariant law is not π_β and the number cannot be interpreted as evidence of correct equilibrium sampling.

(iii) *The score correction substantially reduces the raw-jump distortion.* In the double-well and MoG40 full-law runs, raw CP and LSC-CP share jump streams pathwise. The MB and ϕ^4 comparisons instead match the marginal bank geometry, weights, and intensity but are not pathwise paired. Raw CP retains basin TV 0.309/0.135/0.077 on MoG40/MB/ ϕ^4 . Its terminal mass outside the reference free-energy grid is 31.6%/17.2%/20.9% on those systems, versus $\leq 0.62\%$ for LSC-CP.

Method	W_2	TV_{basin}	$RMSE_{F/(k_B T)}$	$D_{\text{KL}}(p^* \parallel \hat{p})$	FES outside
<i>Double well</i> (reference levels: W_2 0.061 ± 0.033 , TV 0.0032 ± 0.0017)					
ULA	1.26 (0.003)	0.471 (0.001)	1.63 (0.06)	1.09 (0.02)	0
MALA	1.26 (0.003)	0.473 (0.001)	1.63 (0.04)	1.12 (0.02)	0
BAOAB	1.29 (0.002)	0.488 (0.001)	1.99 (0.07)	1.54 (0.04)	0
FLA	0.096 (0.018)	0.0033 (0.003)	1.32 (0.03)	3.4×10^{-5}	0.0033
PT	0.064 (0.032)	0.0033 (0.002)	0.393 (0.089)	3.2×10^{-5}	0
CP	0.176 (0.013)	0.0036 (0.003)	1.58 (0.02)	4.2×10^{-5}	0.029
LSC-CP	0.111 (0.017)	0.0036 (0.003)	1.21 (0.04)	4.4×10^{-5}	0.0047
LSC-CP-RA	0.131 (0.017)	0.0031 (0.003)	1.33 (0.06)	3.2×10^{-5}	0.0084
<i>MoG40</i> (reference levels: W_2 1.31 ± 0.28 , TV 0.051 ± 0.006)					
ULA	26.1 (0.02)	0.916 (0.003)	1.917 (0.008)	4.20 (0.02)	0
MALA	26.1 (0.03)	0.917 (0.003)	1.917 (0.010)	4.21 (0.014)	0
BAOAB	26.2 (0.03)	0.930 (0.004)	1.908 (0.007)	4.24 (0.016)	0
FLA	15.25 (0.21)	0.436 (0.007)	1.481 (0.057)	0.743 (0.045)	0.0011
PT	6.54 (0.34)	0.141 (0.007)	0.574 (0.034)	0.055 (0.005)	0
CP	11.34 (0.18)	0.309 (0.007)	1.761 (0.034)	0.319 (0.020)	0.316
LSC-CP	2.05 (0.29)	0.053 (0.006)	0.695 (0.043)	0.0089 (0.0020)	0.0023
LSC-CP-RA	2.59 (0.25)	0.111 (0.006)	0.888 (0.026)	0.0448 (0.0068)	0.0085
<i>Depth-balanced MB, 10D</i> (reference levels: W_2 0.050 ± 0.020 , TV 0.011 ± 0.007)					
ULA	0.409 (0.003)	0.307 (0.002)	1.961 (0.091)	1.17 (0.14)	0
MALA	0.411 (0.003)	0.308 (0.001)	1.966 (0.083)	1.22 (0.16)	0
BAOAB	0.415 (0.003)	0.309 (0.001)	2.095 (0.071)	1.52 (0.21)	0
FLA	0.179 (0.006)	0.119 (0.008)	1.743 (0.026)	0.048 (0.007)	0.039
PT	0.056 (0.021)	0.013 (0.008)	0.616 (0.036)	5.7×10^{-4}	0
CP	0.353 (0.008)	0.135 (0.006)	1.654 (0.024)	0.047 (0.005)	0.172
LSC-CP	0.050 (0.010)	0.017 (0.009)	1.070 (0.059)	0.00105 (0.00108)	0.0062
LSC-CP-MA	0.047 (0.009)	0.014 (0.006)	1.053 (0.053)	5.8×10^{-4}	0.0062
<i>Coupled ϕ^4, 24D</i> (reference levels: W_2 0.130 ± 0.035 , TV 0.024 ± 0.010)					
ULA	1.22 (0.004)	0.671 (0.003)	2.419 (0.082)	3.07 (0.31)	0
MALA	1.22 (0.004)	0.673 (0.001)	2.449 (0.041)	3.13 (0.17)	0
BAOAB	1.22 (0.005)	0.673 (0.002)	2.452 (0.045)	3.14 (0.22)	0
FLA	0.194 (0.032)	0.060 (0.013)	1.124 (0.037)	0.011 (0.004)	6.3×10^{-5}
PT	0.364 (0.028)	0.119 (0.015)	1.063 (0.056)	0.039 (0.008)	0
CP	0.410 (0.021)	0.077 (0.016)	2.085 (0.027)	0.017 (0.006)	0.209
LSC-CP	0.124 (0.031)	0.028 (0.013)	1.321 (0.126)	0.0023 (0.0018)	0.0036
LSC-CP-MA	0.100 (0.028)	0.022 (0.011)	1.294 (0.088)	0.0015 (0.0013)	0.0029

Table 2: Terminal metrics at $t = T$, mean with SD shown in parentheses where available over seed blocks; entries without parentheses are rounded means. W_2 : sliced (exact in 1D) Wasserstein-2 against the reference in metric space; TV_{basin} : total variation of basin occupancy; $RMSE_{F/(k_B T)}$: dimensionless additive-aligned reduced-free-energy-surface RMSE (numerically an energy error in $k_B T$) on the two-dimensional collective variables (one-dimensional for the double well); D_{KL} : target-to-empirical basin KL with Jeffreys pseudocount; FES outside: empirical mass outside the frozen reference FES grid. Metric definitions in Appendix C.

Raw CP also contacts the finite box on 5.1%, 6.7%, and 0.4% of applied jumps on these systems (Appendix E); this is a numerical-box confound, especially for MoG40 and MB, rather than an intrinsic component of the stationary defect. In one ϕ^4 sensitivity comparison, widening the box from $[-2, 2]^{24}$ to $[-5, 5]^{24}$ changes raw-CP W_2 from 0.28 to 0.41, whereas LSC-CP-MA changes from 0.099 to 0.100. This single check supports robustness to that particular box change but does not establish general box-size independence.

(iv) *Random-bank and deterministic estimators are qualitatively consistent.* Where both are run (double well and MoG40), the paired single-atom estimator preserves the qualitative method ordering and gives comparable terminal errors on several metrics. It is not numerically equivalent on every metric: for example, its MoG40 basin TV is 0.111 versus 0.053, its KL is 0.0448 versus 0.0089, and its outside-grid mass is about four times larger. In MB and ϕ^4 , deterministic and paired multi-atom LSC estimates have similar terminal W_2 , TV, KL, and FES values, but both implementations retain energy discrepancies large relative to their chain MCSEs. These comparisons provide estimator-sensitivity checks, not proofs of numerical equivalence.

5 Post-settling time-series diagnostics

Terminal ensemble metrics quantify the law at time T ; they do not measure the precision of time averages. For that purpose, small independent chains (4 seeds $\times$ 8 chains, 1000 uniformly spaced draws over one additional horizon) are initialized from the system reference, evolved for one full settling horizon, and then analyzed for basin indicators, energy, and collective variables. None is claimed to be an exact stationary draw of the discrete kernel, so the values below are post-settling diagnostics of the numerical chains (Appendix C). A constant basin indicator contributes zero ESS by construction. Raw CP and FLA are excluded because their invariant laws are not π_β .

Method	worst-basin ESS (of 32,000)				energy ESS / signed energy bias				ϕ^4
	DW	MoG40	MB	ϕ^4	DW	MoG40	MB		
ULA	0	0	0	0	14,815 / +0.0003	707 / +0.001	22,223 / +0.007	17,212 / +0.133	
MALA	0	0	0	0	14,869 / +0.0005	710 / -0.004	21,994 / -0.001	15,250 / -0.010	
BAOAB	0	0	0	0	3,658 / -0.0002	534 / -0.001	5,622 / +0.004	2,734 / +0.028	
PT	1,211	572	2,336	2,567	20,494 / -0.001	1,340 / +0.0006	25,202 / -0.0004	24,383 / -0.002	
LSC-CP	1,648	271	2,066	1,222	31,891 / +0.194	24,051 / +0.166	27,993 / +0.040	30,475 / +0.255	
LSC-CP-RA	1,809	859	—	—	31,224 / +0.247	8,119 / +0.717	—	—	
LSC-CP-MA	—	—	1,952	1,150	—	—	26,720 / +0.039	30,972 / +0.251	

Table 3: Post-settling trace diagnostics aggregated over all seeds and chains. Worst-basin ESS is the minimum, over basins, of the summed per-chain effective sample size of that basin’s indicator (Geyer IAT; constant chains contribute zero). Energy bias is the signed difference between the trace mean of V and the reference target, in energy units; it is interpreted with its observable-specific Monte Carlo standard error. The deterministic/numerical LSC-CP row and paired LSC-CP-MA row are shown separately for MB and ϕ^4 . Split- $\hat{R}$ and MCSE values for every observable are reported in the supporting data.

The deterministic/numerical LSC energy MCSEs are 0.0194 (double well), 0.00593 (MoG40), 0.00125 (MB), and 0.0148 (ϕ^4); the paired multi-atom values are 0.00143 (MB) and 0.0141 (ϕ^4), and the double-well and MoG40 single-atom values are 0.0273 and 0.0235. Energy split- $\hat{R}$ is below 1.11 for the tabulated methods, but this does not extend to every basin or collective-variable trace. For example, maximum values are 1.46 for MoG40 LSC-CP, 1.33 for MoG40 LSC-CP-RA, 1.28 for MoG40 PT, and 1.13 for ϕ^4 PT. We therefore use these quantities as finite-horizon diagnostics and

interpret them jointly with population errors.

Three conclusions follow from Table 3.

(i) *Between-basin ESS separates the methods qualitatively.* The three local integrators record *exactly zero* worst-basin ESS on every system because at least one basin indicator is constant or unvisited in the recorded traces. This is not equivalent to zero transitions: on MoG40, for example, ULA, MALA, and BAOAB record 222, 230, and 60 label changes, and the corresponding MB counts are 209, 194, and 43. Their large energy ESS values illustrate why energy decorrelation alone is insufficient evidence of global mixing. LSC-CP sustains worst-basin ESS between 271 and 2,066 for the deterministic/numerical implementation and between 1,150 and 1,952 for the higher-dimensional multi-atom implementation—the same order as fully costed parallel tempering—using local gradients and the jump-bank score.

(ii) *Mixing and accuracy must be read together.* On MoG40, the worst-basin ESS is 572 for PT, 271 for numerical LSC-CP, and 859 for the single-atom estimator; on ϕ^4 it is 2,567 for PT, 1,222 for deterministic LSC-CP, and 1,150 for the multi-atom estimator. The LSC implementations nevertheless have smaller terminal basin KL at the reported horizon (Table 2). The ϕ^4 population difference is reference-sensitive as discussed above. Neither diagnostic alone ranks the samplers; population error, observable bias, mixing, and computational cost must be reported together.

(iii) *The LSC chains carry a material energy discrepancy.* The post-settling energy mean of the deterministic and multi-atom LSC chains exceeds the frozen target estimate by +0.039 to +0.255 energy units; these differences are 17–32 times the reported chain-mean MCSE. The MCSE does not propagate uncertainty in the numerical reference energy, so this comparison is conditional on the frozen reference estimate and is not a complete significance calculation. The single-atom estimator has discrepancies +0.247 (double well) and +0.717 (MoG40), so random-bank pairing or randomization does not by itself control invariant-measure bias. Local methods also show nonzero finite-run differences from the frozen reference estimate: on ϕ^4 , these are +0.1334 for ULA, −0.0100 for MALA, and +0.0276 for BAOAB. In particular, the MALA value should not be attributed to transition-kernel bias without propagating chain, reference, and finite-box uncertainty. The LSC discrepancy cannot be attributed merely to high-energy post-jump chord configurations, because those transients also occur in the exact continuous-time process whose drift–jump balance preserves π_β . Instead, it indicates observable-dependent invariant-measure error in the finite-step kernel, potentially from Euler splitting, drift taming, finite quadrature, exponent capping, or box projection. These devices are useful for robust computation, but systematic refinement is needed to separate their contributions to the measured bias.

Cost-normalized throughput. Using the wall-clock ledger in the current stationarity artifacts, MB energy ESS/s is 14.10 for PT, 5.95 for deterministic LSC-CP, and 6.97 for LSC-CP-MA; the corresponding worst-basin values are 1.307, 0.439, and 0.509. On ϕ^4 , energy ESS/s is 10.67/5.14/6.28 and worst-basin ESS/s is 1.124/0.206/0.233 in the same method order. Per particle step, deterministic LSC uses $Aq_{\rho q\theta} = 128/1024$ score-chord energy differences on MB/ ϕ^4 , whereas the paired multi-atom estimator uses $Aq_\theta = 32/256$. Per-gradient, per-potential, and per-score-quadrature ESS are stored separately; the wall-clock values are single-run, hardware-dependent measurements rather than uncertainty-qualified performance constants (Appendix F).

6 Scope and limitations

Three boundaries of the present evidence are stated explicitly. First, all results are equilibrium-sampling diagnostics of algorithmic chains; passage or round-trip statistics, where recorded, are Kaplan–Meier restricted means of algorithmic mixing times and carry no molecular-kinetics

meaning. Second, the finite-step LSC chains are not exactly π_β -invariant, and the measured consequences—the energy bias of Section 5 and the clipping/boundary diagnostics of Appendix E—are part of the reported results. Third, the present comparison does not include a matched-length *random-geometry* jump control (same atom count, length multiset, weights, rate, and cost; randomized directions) that would isolate the contribution of the designed bank geometry from the generic value of long jumps; raw CP controls for the score correction, not for geometry. Box sensitivity has been checked only for one ϕ^4 change ($L = 2$ to $L = 5$); broader checks at $L = 5, 6, 7$ are needed. Finally, the SNIS and long-PT ϕ^4 references differ by basin TV 0.042, so population-error distinctions below that scale are reference sensitive. The legacy sample-based FES plotting helper also computes $-\log p/\beta$ while labelling the color scale in $k_B T$; those saved FES profile/map color bars are compressed by a factor β and should be regenerated before quantitative use. This plotting defect does not affect the tabulated augmented-grid FES RMSE, which is computed independently from $-\log p$.

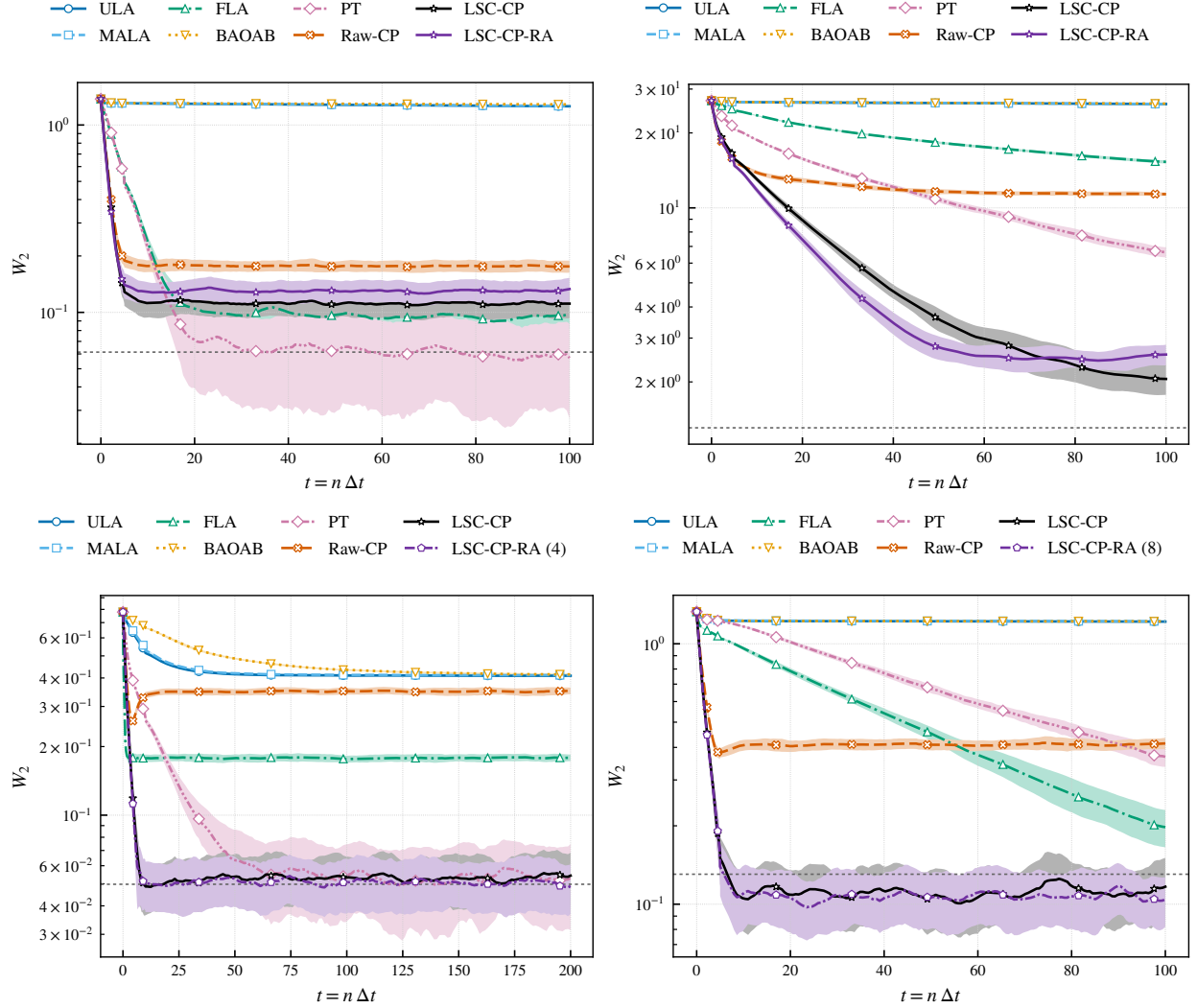


Figure 1: Ensemble relaxation: W_2 to the reference versus simulation time t from the shared localized initial ensembles. Curves and shaded envelopes are centered nine-checkpoint running averages of the seed-block mean and mean $\pm$ one seed-block SD; the dashed horizontal line is the reference mean, not the $\mu_{\text{ref}} + 3s_{\text{ref}}$ descriptive upper band. Top: double well, MoG40; bottom: Müller-Brown 10D, coupled ϕ^4 .

A The Lévy score and the paired-bank estimators

For a finite jump measure ν , the jump generator is $(J_\nu f)(x) = \int [f(x+r) - f(x)] \nu(dr)$, and the Lévy score is

$$S_{\nu,\beta}(x) = - \int_{\mathbb{R}^d \setminus \{0\}} r \int_0^1 \frac{\pi_\beta(x - \theta r)}{\pi_\beta(x)} d\theta \nu(dr) = - \int r \int_0^1 e^{\beta[V(x) - V(x - \theta r)]} d\theta \nu(dr). \quad (7)$$

The chord fundamental theorem of calculus gives, for smooth compactly supported f ,

$$\int S_{\nu,\beta}(x) \cdot \nabla f(x) \pi_\beta(x) dx = - \int (J_\nu f)(y) \pi_\beta(y) dy, \quad (8)$$

hence $\int A_{\nu,\beta} f \pi_\beta dx = 0$ for the generator (2): an infinitesimal, continuous-time weak stationarity statement.

For a finite bank $\nu_{\mathcal{R}}(dr) = \lambda \sum_{a=1}^A w_a \delta_{R_a}(dr)$ the score is

$$S_{\mathcal{R},\beta}(x) = -\lambda \sum_{a=1}^A w_a R_a I_a(x), \quad I_a(x) = \int_0^1 e^{\beta[V(x) - V(x - \theta R_a)]} d\theta, \quad (9)$$

with the chord integral evaluated by Gauss–Legendre quadrature. For the deterministic shell calculation, radial quadrature additionally replaces the continuous shell law by a finite quadrature measure; the full-law jump step still samples the continuous shell. This is a separate approximation from time discretization. The reported MoG40 artifact applies Gauss–Legendre quadrature in chord and radius and a periodic trapezoidal rule in angle. With $u_\ell = (\cos(2\pi\ell/m_\phi), \sin(2\pi\ell/m_\phi))$, normalized radial weights $\hat{w}_j^\rho$, and normalized chord weights $\hat{w}_p^\theta$, its deployed score is

$$S_Q(x) = -\frac{\lambda}{m_\phi} \sum_{\ell=0}^{m_\phi-1} \sum_{j=1}^{q_\rho} \hat{w}_j^\rho \rho_j u_\ell \sum_{p=1}^{q_\theta} \hat{w}_p^\theta e^{\beta[V(x) - V(x - \theta_p \rho_j u_\ell)]}. \quad (10)$$

The saved artifacts use $(q_\theta, q_\rho, m_\phi) = (16, 4, 16)$; the current builder default is $(16, 4, 64)$ for a future rerun. A closed-form chord–radius **MoG40Score** remains in the source tree as a validation comparator, but it is not the deployed production estimator.

The score is accumulated in log space. For each particle, the largest log magnitude is extracted over the positive quadrature/atom contributions *before* their signed vector sum is formed. Rescaling by this common maximum preserves cancellation among opposite vectors. The implementation then replaces the common magnitude $\exp(M)$ by $\exp[\min(M, M_{\max})]$, with $M_{\max} = 600$. When active, this cap preserves the relative atom contributions and the score direction but changes its overall magnitude. It is therefore a numerical-stability device, not an algebraically exact score evaluation. Its activation frequency and the maximum pre-cap exponent are recorded in Appendix E.

A.1 Paired random banks

When the intended law is a mixture of conditional atom laws, $\nu(dr) = \lambda \sum_a w_a q_a(dr)$, the paired multi-atom estimator draws, at the start of each step and independently of the state,

$$R_{n,a} \sim q_a, \quad \hat{\nu}_n(dr) = \lambda \sum_{a=1}^A w_a \delta_{R_{n,a}}(dr), \quad \mathbb{E}[\hat{\nu}_n] = \nu, \quad (11)$$

evaluates a finite- q_θ approximation to (9) for the *realized* bank, and draws the jump counts conditionally on that same bank,

$$N_{n,a} \sim \text{Poisson}(\lambda w_a \Delta t), \quad \Delta J_n = \sum_{a=1}^A N_{n,a} R_{n,a}. \quad (12)$$

With the exact chord integral, each frozen bank would satisfy the weak drift–jump identity (8). The finite chord rule makes this identity approximate, but pairing still removes the additional mismatch that would arise if the score and jumps used independent banks. Averaging over the atom draw makes the estimator unbiased for the *quadrature-defined* score, not for the exact continuous-score integral. The single-atom variant LSC-CP-RA is the $A = 1$ case with the atom family resampled each step.

The implemented step, with integration box $\mathcal{B}$, is

$$\begin{aligned} X_{n+1} = \text{clip}_{\mathcal{B}} \Big[& X_n + \Delta t \frac{b_n}{1 + \Delta t \|b_n\|/c} + \sqrt{2\Delta t/\beta} \xi_n \\ & + \sum_a N_{n,a} R_{n,a} \Big], \\ b_n = & -\nabla V(X_n) + S_{\mathcal{R}_n, \beta}(X_n). \end{aligned} \quad (13)$$

The deployed taming scales are $c = 1$ in E1–E2 and $c = 2h$ in E3–E4, where h is the shell half-width. Euler splitting, finite quadrature, exponent capping, drift taming, and box projection all lie outside the exact generator identity. The weak-identity residual tests the uncapped quadrature-level drift–jump cancellation only; it does not test these finite-step modifications. The atomwise counts in (12) are untruncated. The separate full-law compound-Poisson implementation uses a safety cap $K_{\max} = 8$ on applied multiplicity while recording the uncapped Poisson draw; no cap hit occurred in the reported trajectories. Multiple events for one paired atom within a step reuse its stored displacement, an $O(\Delta t^2)$ difference from independently resampling every full-law event.

B System specifications

Double well. $V(x) = (x^2 - 1)^2$, $\beta = 8$; bank: atoms ± 2 , weights $1/2$, radial half-width 0.2 ; box $[-5.2, 5.2]$ (widened so that raw CP’s excursions do not pile mass at the edge of the density grid).

MoG40. Means μ_k drawn once (fixed seed) in $[-40, 40]^2$; $\pi_\beta = \frac{1}{40} \sum_k N(\mu_k, I_2)$; annulus law with radius uniform on $[10, 14]$, direction uniform; box $[-65, 65]^2$. The artifact/provenance qualification in Section 2 applies.

Depth-balanced Müller–Brown. Latent potential

$$V_{\text{MB3}}(z) = \sum_{k=1}^4 D_k \exp\{a_k(z_1 - x_k)^2 + b_k(z_1 - x_k)(z_2 - y_k) + c_k(z_2 - y_k)^2\}, \quad (14)$$

k	D_k	a_k	b_k	c_k	x_k	y_k
1	−1.6607	−1	0	−10	1	0
2	−1.0000	−1	0	−10	0	0.5
3	−1.0218	−6.5	11	−6.5	−0.5	1.5
4	0.1500	0.7	0.6	0.7	−1	1

(depths retuned so all three minima sit at $V \simeq -0.7957$). With $x = zB^\top$, the full ten-dimensional potential is

$$V_3(x) = V_{\text{MB3}}(z_1, z_2) + \frac{1}{2(0.4)^2} \sum_{j=3}^{10} z_j^2, \quad z = xB^{-\top}. \quad (15)$$

Here $B = Q \text{diag}(0.75, \dots, 1.45)$, with ten equally spaced diagonal entries and Q obtained by QR factorization of the `default_rng(12345)` Gaussian matrix. The inverse temperature is $\beta = 24$. Bank: four directed relay atoms $A \rightarrow B$, $B \rightarrow A$, $B \rightarrow C$, $C \rightarrow B$, weights $1/4$, radial half-width $0.1 \min_a \|r_a\|$; drift cap twice that half-width.

Coupled ϕ^4 . Site potential

$$W(u, v) = (u^2 - 1)^2 + (v^2 - 1)^2 - 0.0125 uv + 0.0075 u + 0.015 v, \quad (16)$$

$\kappa = 2.5$, $\delta = 1/12$, twelve periodic sites with $q_{13} = q_1$, $\beta = 8$. Bank: eight coherent edge atoms $\mathbf{1}_{12} \otimes (v_j - v_i)$ along the phase square, weights $1/8$, radial half-width $0.1 \min_a \|r_a\|$, drift cap twice that half-width. The integration box half-width $L = 5$ is *derived*: a simultaneous Laplace-component envelope with union-bound tail budget $\alpha = 10^{-8}$ gives $B_\beta = 1.910$, the hottest-PT envelope $B_{\beta_{\min}=1} = 3.569$, and the maximum componentwise one-jump reach $R_\infty = 2.200$; $L = \lceil \max\{B_\beta + R_\infty, B_{\beta_{\min}}\} \rceil = 5$. The basin map for $\bar{q}$ is constructed by gradient flow on $[-4, 4]^2$ —the *jump-reachable* order-parameter set (double-jump reach $|\bar{q}| \leq 3.3$), not merely the phase minima (Appendix E).

Parallel-tempering ladders (auto-tuned, fully costed). Double well: $\beta \in \{8, 1\}$ ($K = 2$, swap acceptance 0.41); MoG40: $\{8, 0.447, 0.025\}$ ($K = 3$, 0.26); MB: $\{24, 13.5, 7.6, 4.3\}$ ($K = 4$, 0.37); ϕ^4 : six geometric rungs from 8 to 1 ($K = 6$, 0.31).

Baseline implementation details. ULA, MALA, FLA, and BAOAB use the same finite state box as the corresponding experiment and tame their local drift or force with $c = 1$. MALA uses $h = 2\Delta t/\beta$ and rejects, rather than clips, proposals outside the box. BAOAB uses unit mass and friction $\gamma = 1$. FLA uses independent coordinatewise symmetric α -stable increments with $\alpha = 1.7$. Parallel tempering performs MALA within each replica and alternates adjacent swap pairs every ten steps; reported distributional metrics use the cold replica, while all replica evaluations and wall time are charged.

C Metric definitions

Evaluation coordinates and Wasserstein distance. Let $\xi(x)$ denote the coordinate in which a distributional metric is evaluated. We use $\xi(x) = x$ for E1 and E2, the two active coordinates $\xi(x) = (z_1, z_2)$ for E3, and $\xi(q) = \bar{q} = 12^{-1} \sum_{i=1}^{12} q_i$ for E4. For equal-size empirical samples in one dimension (E1), the exact empirical Wasserstein distance is

$$\widehat{W}_2 = \left[\frac{1}{N} \sum_{i=1}^N (X_{(i)} - Y_{(i)})^2 \right]^{1/2}. \quad (17)$$

For E2–E4, the table entry labelled W_2 is the sliced distance

$$\widehat{\text{SW}}_2 = \left[\frac{1}{L} \sum_{\ell=1}^L W_2^2 \left((\theta_\ell^\top)_\# \widehat{\mu}_N, (\theta_\ell^\top)_\# \widehat{\nu}_N \right) \right]^{1/2}, \quad L = 200. \quad (18)$$

The unit vectors θ_ℓ are drawn once with seed 777 and reused across methods and times. E3 additionally records a ten-dimensional sliced W_2 with a separate frozen projection set (seed 778); it is not the W_2 shown in Table 2. A 500-point Hungarian matching is only a terminal two-dimensional spot check.

Maximum mean discrepancy. The reported MMD is the square root of the biased Gaussian-kernel V-statistic,

$$\widehat{\text{MMD}} = \left[\frac{1}{N^2} \sum_{i,j} k(X_i, X_j) + \frac{1}{N^2} \sum_{i,j} k(Y_i, Y_j) - \frac{2}{N^2} \sum_{i,j} k(X_i, Y_j) \right]^{1/2}. \quad (19)$$

$$k(x, y) = e^{-\|x-y\|^2/(2\sigma^2)}. \quad (20)$$

The diagonal terms are retained. The bandwidth σ is the median pairwise distance among at most 2,048 frozen reference points (seed 99) and is not re-estimated along the trajectory.

Secondary density, energy, and Stein diagnostics. For E1, the density TV uses 350 fixed bins on $[-5.2, 5.2]$:

$$\text{TV}_{\text{density}} = \frac{1}{2} \sum_{b=1}^{350} |\hat{p}_b - p_b^*|. \quad (21)$$

This differs from the two-well occupancy TV. The energy-overlap deficit uses 40 frozen energy bins,

$$e_{\text{overlap}} = 1 - \sum_b \min\{\hat{h}_b, h_b^*\}, \quad (22)$$

with empirical energy outside the reference grid retained as nonoverlap. The archived kernel Stein discrepancy is the square root of the biased V-statistic

$$\widehat{\text{KSD}} = \left[\frac{1}{m^2} \sum_{i,j=1}^m (\mathcal{T}_\pi^x \mathcal{T}_\pi^y k)(X_i, X_j) \right]^{1/2}, \quad k(x, y) = (1 + \|x - y\|^2)^{-1/2}, \quad (23)$$

where the score is $\nabla \log \pi_\beta = -\beta \nabla V$, $m \leq 512$, and a fixed seed 17 chooses the subsample. KSD can be small for a chain trapped in one mode and is therefore secondary to the basin-aware metrics. Additional first-CV CDF/KDE discrepancies are archived for collaborator parity but do not enter the principal claims.

Basin occupancy, TV, and KL. For basin partition $\{\Omega_k\}_{k=1}^K$ with target masses p^* , $\hat{p}_k = \frac{1}{N} \sum_i \mathbf{1}\{X_i \in \Omega_k\}$, $\text{TV}_{\text{basin}} = \frac{1}{2} \sum_k |\hat{p}_k - p_k^*|$, and

$$D_{\text{KL}}(p^* \|\tilde{p}) = \sum_{k:p_k^* > 0} p_k^* \log \frac{p_k^*}{\tilde{p}_k}, \quad \tilde{p}_k = \frac{\hat{p}_k + \alpha/N}{1 + K\alpha/N}, \quad \alpha = \frac{1}{2}. \quad (24)$$

The target-to-empirical orientation penalizes a sampler that misses a target basin; the Jeffreys-scale pseudocount keeps the statistic finite. Basin TV is the total variation of this finite partition and therefore only a lower bound on full-distribution TV. In E2, $p_k^* = 1/40$ is the operational nearest-center target used by the implementation; it is not asserted to equal the exact Gaussian-mixture probability of every finite Voronoi cell.

The actual partitions are system specific. E1 uses the sign of x ; E2 uses the nearest frozen center. E3 constructs a 600^2 gradient-flow label map on $[-2, 1.9] \times [-1.3, 2.6]$ in (z_1, z_2) , with flow step 1.5×10^{-4} for 40,000 steps, and integrates target basin masses on a 1200^2 grid. E4 constructs an 800^2 gradient-flow label map of the site potential W on $[-4, 4]^2$ in the $\bar{q}$ plane, using the same flow step and count; its target masses are the direct-SNIS weighted category probabilities. The sign-phase grouping used only to condition the E4 visualization KDE is not this gradient-flow basin partition.

Explored-mode coverage. For the uniform target masses in E1 and E2,

$$\text{EMC} = \frac{\exp[H(\hat{p})]}{K}, \quad H(\hat{p}) = - \sum_k \hat{p}_k \log \hat{p}_k. \quad (25)$$

For the nonuniform E3 and E4 target masses, the implementation instead uses

$$\text{EMC} = 1 - \text{JS}_2(\hat{p}, p^\star), \quad (26)$$

where JS_2 is the base-two Jensen–Shannon divergence. Both definitions have optimum 1, but their numerical values are not comparable across the two cases.

Augmented-grid free-energy RMSE. The current implementation uses a discrete augmented grid, not the earlier $5/N$ reference-weighted definition. Let J be the number of rectangular interior cells and let cell $J+1$ collect all mass outside the frozen grid. With $\alpha = 1/2$, the smoothed empirical and reference probabilities are

$$\hat{p}_i = \frac{n_i + \alpha}{N + (J+1)\alpha}, \quad p_i^{\text{ref}} = \frac{n_i^{\text{ref}} + \alpha}{M + (J+1)\alpha}. \quad (27)$$

For E4, $n_i^{\text{ref}} = M \sum_{r: \xi(Q_r) \in C_i} w_r$ is the direct-SNIS effective count; all importance weights are first rescaled to sum to M . The retained set and reduced free-energy difference are

$$\mathcal{I} = \{i : p_i^{\text{ref}} \geq 1/N\} \cup \{J+1\}, \quad \Delta_i = -\log \hat{p}_i + \log p_i^{\text{ref}}. \quad (28)$$

The off-grid cell is retained even though its only reference mass may arise from the pseudocount. Uniform cell weights are then used:

$$\text{RMSE}_{F/k_B T} = \left[\frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (\Delta_i - c^\star)^2 \right]^{1/2}, \quad c^\star = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \Delta_i. \quad (29)$$

This is a dimensionless reduced-free-energy error, numerically equal to an energy error measured in units of $k_B T$. For equal-volume interior cells, $-\log p_i$ and $-\log(p_i/|C_i|)$ differ only by the fitted additive constant; the off-grid cell is a diagnostic discrete state rather than a physical volume element. The raw, unsmoothed empirical off-grid probability is additionally reported as FES outside.

The grids contain 40 bins in E1 and 15^2 , 14^2 , and 10^2 cells in E2–E4. Each coordinate range is the reference minimum–maximum interval expanded by 5%. E1–E3 use 50,000 reference points for the FES histogram; E4 uses the 200,000 weighted production SNIS proposals. Because the same half-count is combined with different empirical and reference sample totals, and because the off-grid cell is always retained, the metric includes a sample-size-dependent smoothing contribution even when raw outside mass is zero. No finite-sample reference band is currently available for this FES metric.

Energy errors and signed time-series bias. The terminal ensemble reports

$$e_{\mu_V} = \left| \frac{1}{N} \sum_i V(X_i) - \mu_V^* \right|, \quad e_{\sigma_V^2} = \left| \frac{1}{N-1} \sum_i (V(X_i) - \bar{V})^2 - (\sigma_V^2)^* \right|. \quad (30)$$

These are absolute errors. In contrast, a post-settling observable row uses the signed bias $\bar{O} - O^*$. E4 reference moments are direct-SNIS weighted moments; E1–E3 use the frozen numerical reference sample.

IAT, ESS, MCSE, $\hat{R}$. Per chain, the integrated autocorrelation time uses Geyer’s initial-positive/initial-monotone sequence. Writing $\Gamma_k = \hat{\rho}_{2k} + \hat{\rho}_{2k+1}$, pair sums are truncated before the first nonpositive value and replaced by their cumulative minimum $\tilde{\Gamma}_k$:

$$\widehat{\tau}_{\text{int}} = \max \left\{ 1, -1 + 2 \sum_k \tilde{\Gamma}_k \right\}. \quad (31)$$

A constant chain is assigned $\tau_{\text{int}} = \infty$ and zero ESS. Independent chains contribute

$$\text{ESS}_{\text{total}} = \sum_m \frac{n_m}{\widehat{\tau}_{\text{int}m}}, \quad \text{MCSE} = \frac{s_{\text{pooled}}}{\sqrt{\text{ESS}_{\text{total}}}}. \quad (32)$$

This MCSE quantifies uncertainty of the correlated chain mean only; it does not include uncertainty in the separately estimated reference target O^* . Thus total draws divided by an arithmetic mean IAT is not used. Split- $\hat{R}$ is rank normalized with average ranks for ties and Blom transform $(r - 3/8)/(S + 1/4)$; the reported value is the maximum of the bulk and folded statistics. Globally constant traces give an undefined $\hat{R}$. Neither a large ESS nor $\hat{R}$ close to one proves that the numerical kernel targets π_β .

Finite-sample reference levels and threshold time. Twenty replicates estimate the mean μ_{ref} and sample standard deviation s_{ref} of reference–reference W_2 /MMD (and E3 ten-dimensional W_2), or of a fresh reference sample against the target for basin TV, EMC, and E1 density TV. For lower-is-better errors the descriptive band is $\mu_{\text{ref}} + 3s_{\text{ref}}$; it is not a mathematical lower bound. The saved `time_to_threshold_TV` is separately defined as the first checkpoint at which the seed-mean TV is no larger than $2\mu_{\text{ref}}$ for five consecutive checkpoints. It does not use the $\mu + 3s$ band and should not be called a convergence or mixing time.

Cost accounting. The aggregate function-evaluation ledger is

$$\text{NFE} = n_V + n_{\nabla V} + n_{\Delta V}, \quad (33)$$

where $n_{\Delta V}$ counts score-chord potential differences. These three evaluations have different arithmetic and memory costs, so the aggregate is only a bookkeeping convention. PT includes all replicas; reference generation, diagnostics, plotting, and I/O are excluded. Main-ensemble seeds are batched, so their wall time and NFE are batch totals repeated in seed rows, and a zero reported wall-time standard deviation is not timing uncertainty. Stationarity ESS/s uses separately synchronized sampler-only timings.

Passage statistics. Where passage times are recorded they are right-censored at the common horizon, restricted to chains starting in the declared home core, and summarized by the Kaplan–Meier restricted mean; the model-dependent censored-exponential estimate, when saved, is labelled as such. Both are algorithmic-mixing diagnostics only.

D References and the ϕ^4 importance reference

E1–E3. The double-well reference is a dense-grid inverse CDF on the declared box (2×10^5 grid points), and MoG40 permits exact mixture draws. In E3, writing $z = B^{-1}x$, the target factorizes as

$$\pi_z(z) \propto e^{-24V_{\text{MB3}}(z_1, z_2)} \prod_{j=3}^{10} \mathcal{N}\left(z_j; 0, \frac{\sigma_{\text{aux}}^2}{24}\right), \quad \sigma_{\text{aux}} = 0.4. \quad (34)$$

The reference therefore uses a 2400^2 numerical grid for the active plane and exact Gaussian auxiliary draws. The E3 density panel evaluates and normalizes $e^{-24V_{\text{MB3}}(z_1, z_2)}$ directly on the display grid; it is not a histogram.

E4 production importance reference. The exact marginal of the spatial mean is well defined by

$$p_{\bar{q}}(s) = Z_{\beta}^{-1} \int \delta^{(2)}\left(s - \frac{1}{12} \sum_{i=1}^{12} q_i\right) e^{-\beta V(q)} \prod_{i=1}^{12} dq_i, \quad (35)$$

but it has no simple closed form for the coupled quartic field. The numerical reference uses the four-phase Laplace-mixture proposal

$$q_{\text{prop}}(x) = \sum_{k=1}^4 a_k \mathcal{N}(x; m_k, (\beta H_k)^{-1}), \quad a_k \propto e^{-\beta V(m_k)} (\det H_k)^{-1/2}, \quad (36)$$

where $H_k = \nabla^2 V(m_k)$. For proposals $Q_r \sim q_{\text{prop}}$,

$$\tilde{w}_r = \frac{\exp[-\beta V(Q_r) - \log q_{\text{prop}}(Q_r)]}{\sum_s \exp[-\beta V(Q_s) - \log q_{\text{prop}}(Q_s)]}. \quad (37)$$

The production GPU reference has 2×10^5 proposals, weight ESS 41,182 (20.6%), and maximum normalized weight 3.6×10^{-3} . Basin masses, energy moments and the energy-overlap histogram, CV means, and the FES histogram use these weights directly. The current W_2 and MMD routines accept unweighted point clouds, so those two diagnostics use finite sampling-importance-resampling with oversampling factor 16 and are therefore approximate; this is an implementation choice, not a mathematical restriction on weighted empirical measures.

E4 visualization-only marginal density. The reference panel uses the same model, proposal, proposal count, and nominal seed in an independent CPU reference integration, not the archived CUDA draws. Because CPU and CUDA random streams are backend-specific, its proposal cloud is not bitwise identical: the figure cache has ESS 107,246, whereas the metric reference retains the production ESS 41,182. Let $g(r)$ be the sign phase of $s_r = \bar{q}(Q_r)$, let

$$N_{\text{eff},g} = \left[\sum_{r:g(r)=g} \left(\frac{\tilde{w}_r}{\sum_{s:g(s)=g} \tilde{w}_s} \right)^2 \right]^{-1}, \quad (38)$$

and let $\hat{\Sigma}_g$ be the corresponding weighted covariance. The display estimator is

$$\hat{p}_{\bar{q}}(s) = \sum_{g=1}^4 \sum_{r:g(r)=g} \tilde{w}_r \varphi_2\left(s; s_r, N_{\text{eff},g}^{-1/3} \hat{\Sigma}_g\right). \quad (39)$$

On the display region $[-5, 5]^2$, the implementation imposes

$$\lambda_{\min}(\hat{H}_g) \geq [0.65 \min(\Delta x, \Delta y)]^2 \quad (40)$$

on every bandwidth matrix and evaluates the continuous KDE above by weighted binning and FFT convolution on an 801^2 grid. It then truncates negative roundoff artifacts and renormalizes the density. Phase conditioning prevents a global bandwidth from creating artificial bridges between modes; it is a visualization device and does not replace the production basin map or weighted FES histogram.

E Numerical convergence and stability diagnostics

Diagnostic	DW	MoG40	MB	ϕ^4
nonfinite states	0	0	0	0
score-cap fraction, reported LSC	5.02×10^{-4}	0	2.85×10^{-7}	2.09×10^{-4}
state-box fraction, maximum over methods	8.67×10^{-5}	7.92×10^{-4}	3.304×10^{-3}	8.903×10^{-4}
jump-boundary contact, reported LSC	1.69×10^{-4}	2.13×10^{-5}	2.18×10^{-3}	1.60×10^{-4}
jump-boundary contact, raw CP	3.03×10^{-4}	5.117×10^{-2}	6.733×10^{-2}	4.130×10^{-3}
maximum basin-map outside mass, targeting rows	0	0	0	10^{-3}
uncapped weak-identity residual	1.5×10^{-7}	2.1×10^{-8}	5.0×10^{-7}	2.0×10^{-7}

Table 4: Numerical-stability measurements for the reported trajectories. Score-cap and state-box entries are fractions of particle proposals; the state-box row is the maximum over all compared methods. Jump-boundary contact is measured per applied jump. The basin-map row is the maximum over all targeting-method rows and recorded checkpoints.

The score-cap activation is small but not identically zero for the double well, MB, or ϕ^4 . On the deliberately extended weak-identity grids, the fractions with pre-cap exponent above $M_{\max}$ are 0.402 for the double well and 0.374 for MB; these grid values are distinct from the trajectory activation frequencies in Table 4. The residual itself is formed from the uncapped log-domain score and tests the generator identity for the quadrature-defined measure and a finite family of observables. It does not test the exponent cap, drift taming, Euler splitting, or box projection, and it is not a certificate of the invariant distribution of the stabilized finite-step chain.

Because compound-Poisson jumps occur independently of the current state, rare events can contact any finite computational box. The contact and projection frequencies in Table 4 quantify this exposure. In particular, the larger raw-CP contact rates on MoG40 and MB make the finite box a material confound in that ablation. For ϕ^4 , the basin map was constructed on $[-4, 4]^2$ to cover the jump-reachable order-parameter region rather than only the phase minima. Its targeting-method maximum is 10^{-3} : one of $N = 1000$ particles lies outside the map at the relevant checkpoint for both deterministic LSC-CP and LSC-CP-MA, while the terminal seed-block mean returns to zero (raw CP reaches 5×10^{-3}). Even rare cap, projection, or map-domain events matter for interpretation, so low frequencies are evidence of numerical robustness, not exact stationarity.

F Computational details and data availability

The production manifests record float64 arithmetic on one NVIDIA H200 GPU, Python 3.12.13, and PyTorch 2.13.0+cu130. The repository mirror contains the terminal positions, metric time series, terminal summaries, manifests, and per-observable stationarity summaries. It does not currently

mirror the raw stationarity trace arrays or the original/resolved run configurations named by the absolute paths in the manifests; consequently, ESS and provenance cannot be recomputed from the clone alone. Gradient, potential, and Lévy-score-quadrature evaluation counts are nevertheless stored separately for each reported observable.

The MB and ϕ^4 ESS/s values in Section 5 use the synchronized sampler-only timings stored in the current stationarity summaries. They are single-run measurements and are not accompanied by timing replicates. Source code, complete immutable run artifacts, resolved configurations, and analysis-ready numerical data should be deposited together in the publication archive.